GG - MATH.(B & E)

Q6344888
Question Paper SL No.

2026
MATHEMATICS

Full Marks — { 90 — **For Regular Candidates**
100 — **For External Candidates**

Special credit will be given for answers which are brief and to the point.
Marks will be deducted for spelling mistakes, untidiness, overwriting and bad handwriting.

[The answers of the Question Nos. 1, 2, 3, 4 are to be written at the beginning of the answer-script mentioning the question numbers in the serial order. Necessary calculation and drawing must be given on the right hand side by drawing margins on the first few pages on the answer-script. Tables and Calculators of any type are not allowed. Approximate value of π may be taken as $\frac{22}{7}$, if necessary. Graph paper will be supplied with question paper. Arithmetic problems may be solved by algebraic method.]

[Alternative Question No. 11 is given for Sightless Candidates on Page No. 16]

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1. Choose the correct option in each case from the following questions: $1 \times 6 = 6$

(i) If a principal becomes double in 10 years, the rate of simple interest per annum is

(a) 5% (b) 10%

(c) 15% (d) 20%

(ii) The condition for the two roots of a quadratic equation $ax^2 + bx + c = 0$ $(a > 0)$ to be equal in magnitude but opposite in sign is

(a) $b = 0,\ c = 0$ (b) $b = 0,\ c > 0$

(c) $b = 0,\ c < 0$ (d) $b > 0,\ c = 0$

(iii) If the average of $6, 7, x, y, 16$ is 9, then

(a) $x + y = 21$ (b) $x + y = 16$

(c) $x - y = 21$ (d) $x - y = 19$

(iv) An arc of length 121 cm of a circle makes an angle of 77° at the centre. The radius of the circle is

(a) 110 cm (b) 100 cm

(c) 90 cm (d) 70 cm

(v) If the length of a side of a cube is a units and the length of its diagonal is d units, then the relation between a and d is

(a) $\sqrt{2}a = d$ (b) $\sqrt{3}a = d$

(c) $a = \sqrt{3}d$ (d) $a = \sqrt{2}d$

(vi) $ABCD$ is a cyclic quadrilateral whose centre is O. BC is extended up to E. If $\angle DCE = 96^\circ$, then the value of $\angle BOD$ is

(a) 42° (b) 84°

(c) 142° (d) 168°

2. Fill up the blanks (any **five**): $1 \times 5 = 5$

(i) If the ratio of principal and yearly amount is $8 : 9$, then the yearly rate of interest is

.....

(ii) The conjugate surd of $(\sqrt{3} - 5)$ is .....

(iii) Two tangents at the endpoints of a diameter of a circle are mutually .....

(iv) If $x = a \sec\theta$ and $y = b \cot\theta$, then $\frac{x^2}{a^2} - \frac{b^2}{y^2} =$ .....

(v) If the radius of a solid hemisphere is $3r$ units, its total surface area is .....

(vi) The frequencies of $1, 2, 3, 4, 5$ are respectively $1, 2, 3, 4, f$. If their arithmetic mean is 4, then the value of f is .....

3. Write **True** or **False** (any **five**): $1 \times 5 = 5$

(i) $\sin^2\theta = (\sin\theta)^2$, where $0^\circ < \theta < 90^\circ$.

(ii) The length of a side of the largest cube inscribed in a sphere of radius 4 cm is $4\sqrt{2}$ cm.

(iii) The angle in the segment of a circle which is greater than a semicircle is an obtuse angle.

(iv) If the arithmetic mean of $x-3$, $x-1$, 7, x, $2x-1$, $3x-5$ is 7.5, then their median will be 3.

(v) If $x \propto \frac{1}{y}$, then $(xy)^{10}$ is a constant.

(vi) In a business, the ratio of the capitals of Raju and Asif is $5:4$. If Raju gets Rs. 80 out of the total profit, then Asif gets Rs. 100.

4. Answer any **ten** questions: $2 \times 10 = 20$

(i) A and B started a business by investing Rs. $15,000$ and Rs. $45,000$ respectively. After 6 months, B got a profit of Rs. $3,030$. What is the profit of A?

(ii) In a triangle ABC, a straight line parallel to BC intersects AB and AC at P and Q respectively. If $AP = 4$ cm, $QC = 9$ cm and $PB = AQ$, find the value of PB.

(iii) Two chords AB and CD are equidistant from the centre O of a circle. If $\angle AOB = 60^\circ$ and $CD = 6$ cm, find the radius of the circle.

(iv) If $\tan\theta + \cot\theta = 2$, find the value of $\tan^7\theta + \cot^7\theta$.

(v) If x and y are positive real numbers, is $\sec\theta = \frac{x}{y}$ possible? Give your answer with reason.

(vi) For two right circular cylinders, if the ratio of their heights is $1:2$ and the ratio of the circumferences of their bases is $3:4$, find the ratio of their volumes.

(vii) The arithmetic mean of $x_1, x_2, \ldots, x_n$ is $\bar{x}$. Prove that $\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$.

(viii) If the rate of interest increases from 5.5% to 6%, the yearly interest increases by Rs. 49.50. Find the capital.

(ix) If the sum of the roots of the equation $x^2 - 4x = K(x-1) - 5$ is 7, find the value of K.

(x) If $(a+b) : \sqrt{ab} = 2:1$, find $a:b$.

(xi) If the radius of a sphere is increased by 50%, find the percentage increase in its volume.

(xii) $ABCD$ is a cyclic quadrilateral. If $AD = AB$, $\angle DAC = 60^\circ$ and $\angle BDC = 50^\circ$, find $\angle ACD$.

5. Answer any **one** question: 5

(i) If the rate of compound interest is 4% in the 1st year and 5% in the 2nd year, find the interest on Rs. $25,000$ for two years.

(ii) Three friends invest Rs. $4,800$, Rs. $6,600$ and Rs. $9,600$ respectively in a business. The 1st person receives $\frac{1}{8}$ of the profit as salary for looking after the business, and the remaining profit is distributed among them in the ratio of their capitals. If after one year the 1st person receives Rs. 780, find the amounts received by the other two.

6. Answer any **one** question: 3

(i) Solve: $b(c-a)x^2 + c(a-b)x + a(b-c) = 0$.

(ii) The digit in the tens place of a two-digit number is 3 less than the digit in the units place. The product of the digits is 15 less than the number. Find the number.

7. Answer any **one** question: 3

(i) If $(x^3 + y^3) \propto (x^3 - y^3)$, prove that $(x^2 + y^2) \propto xy$.

(ii) If $x(2-\sqrt{3}) = y(2+\sqrt{3}) = 1$, find the value of $3x^2 - 5xy + 3y^2$.

8. Answer any **one** question: 3

(i) If $\frac{a+b-c}{a+b} = \frac{b+c-a}{b+c} = \frac{c+a-b}{c+a}$ and $a+b+c \neq 0$, prove that $a = b = c$.

(ii) If $x = \frac{8ab}{a+b}$, find the value of $\frac{x+4a}{x-4a} + \frac{x+4b}{x-4b}$.

9. Answer any **one** question: 5

(i) Prove that the angle subtended at the centre of a circle by an arc is double the angle subtended by the same arc at any point on the remaining part of the circle.

(ii) If two circles touch each other, prove that the point of contact lies on the line joining their centres.

10. Answer any **one** question: 3

(i) In the isosceles triangle ABC, $\angle B$ is a right angle. The bisector of $\angle BAC$ intersects BC at D. Prove that $CD^2 = 2BD^2$.

(ii) O is a point inside the rectangle $ABCD$. Prove that $OA^2 + OC^2 = OD^2 + OB^2$.

11. Answer any **one** question: 5

(i) In $\triangle ABC$, the base $BC = 6$ cm, $\angle ABC = 60^\circ$ and $AB = 8$ cm. Draw the circumcircle of the triangle.

(ii) Construct a square equal in area to an equilateral triangle of side 6 cm.

12. Answer any **two** questions: $3 \times 2 = 6$

(i) If the ratio of the three angles of a triangle is $2 : 3 : 4$, determine the circular measure of the greatest angle.

(ii) If $\tan\theta = \frac{4}{3}$, find the value of $\sin\theta + \cos\theta$.

(iii) If A and B are two complementary angles, prove that $(\sin A + \cos B)^2 = 1 + 2\sin A \sin B$.

13. Answer any **one** question: 5

(i) From the roof of a building, the angles of depression of the top and foot of a lamp post are 30° and 60° respectively. If the ratio of the heights of the building and the lamp post is $3 : 2$, find the required angle θ.

(ii) From the foot of a hill, the angle of elevation of its top is 45°. After moving 100 m towards the hill along a slope inclined at 30°, the angle of elevation of the top becomes 60°. Find the height of the hill.

14. Answer any **two** questions: $4 \times 2 = 8$

(i) The ratio of the length, breadth and height of a solid rectangular parallelepiped is $4 : 3 : 2$ and its total surface area is 468 sq. cm. Find its volume.

(ii) The internal and external radii of a hollow cylinder of height 20 cm are 4 cm and 5 cm respectively. By melting this cylinder, a solid cone of height equal to one-third of the height of the cylinder is formed. Find the diameter of the base of the cone.

(iii) A hemispherical bowl of internal radius 9 cm is full of water. How many cylindrical bottles of diameter 3 cm and height 4 cm are required to contain this water?

15. Answer any **two** questions: $4 \times 2 = 8$

(i) Find the arithmetic mean of the following distribution:

Class	$5-14$	$15-24$	$25-34$	$35-44$	$45-54$	$55-64$
Frequency	3	6	18	20	10	3

(ii) Making a cumulative frequency (greater than type) distribution table of the given data, draw an ogive on graph paper:

Class	$100-120$	$120-140$	$140-160$	$160-180$	$180-200$
Frequency	8	14	10	12	4

(iii) Find the mode of the following frequency distribution:

Marks obtained	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
Number of Students	8	20	29	42	60	70

[ALTERNATIVE QUESTION FOR SIGHTLESS CANDIDATES]

11. Answer any **one** question: 5

(i) Describe the process of drawing the circumcircle of a triangle.

(ii) Describe the process of constructing a square equal in area to an equilateral triangle.